

A3-PS015-TC554

Explainable Home and Last-Mile Connectivity Diagnosis

NERVE - ***Network Evidence, Reasoned Verification & Explanation***

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AIORI Hackathon 3
THREE MUSKETEERS

Problem Description

CORE PROBLEM

- Speed tests **only** show symptoms, not the cause: Running a speed test tells you bandwidth is low or ping is high, but never explains ***where or why*** the issue is happening.

The *"Blame Game"* confusion: Everyday users cannot pinpoint which part of their network is failing:

- Is it their own Wi-Fi / local router?
- Is it a DNS issue stopping websites from loading?
- Is it packet loss on a specific path to one website?
- Is it network congestion because someone else is uploading or downloading?

REAL-WORLD USER PAIN POINTS

Work & Study disruption:

- Remote workers face frozen video calls during household file transfers.
- Students see browser errors even though the computer shows "Connected".

Blind, trial-and-error fixes:

- Users waste time randomly restarting routers, swapping cables, or tweaking system settings without knowing if it helps.

SUPPORT & ISP CHALLENGES

- **Vague customer complaints:** Support engineers get tickets saying "internet is slow" or "Wi-Fi is down" instead of measurable evidence.
- **No verifiable logs:** Technicians lack timestamps, probe targets, and path metrics needed to quickly isolate and reproduce the problem.
- **Guesswork over proof:** Most existing diagnostic tools force a speculative guess instead of honestly stating when evidence is missing or inconclusive.

Background



The Problem: Users only know their call froze or a page failed. Traditional speed tests just give a raw Mbps number, with zero clue which part of the path choked.

NERVE → ***Lets user pick the symptom directly to focus initial checks***

Endpoint Blindness

The Problem: device perceives sensations at the edge, not inside the ISP core or remote hardware.

NERVE → ***Reports what was measured, never an unsubstantiated "ISP fault".***

The Problem: Tools run brute-force tests in a fixed order, or use black-box guesses that blame the ISP without proof

NERVE → ***Adaptive policy picks only the next-best probe to separate hypotheses; deterministic rules issue findings or honestly say Inconclusive.***

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The Problem: Users try random reboots with zero follow-up or confirmation whether anything changed.

NERVE → ***Suggests safe, consent-gated fixes and immediately re-tests the matched window to prove if the connection measurably improved.***

THE BIGGER PROBLEM (Brute-Force Scans):

Standard tools run every heavy test in a fixed order regardless of results, wasting user time and data.

NERVE → *Adapts to what it senses—picks only the probe that separates remaining explanations, stops when evidence is sufficient, and returns Inconclusive when it cannot prove the cause.*

literature and RFC review

Key Standards and Their Impact on NERVE's Design

Standard (RFC)	Relevance		NERVE Design Decision
RFC 2308 DNS negative caching	Explains caching of negative DNS responses (NXDOMAIN).	→	Controlled **nerve.lab** zone, so no **NXDOMAIN** noise.
RFC 826 / 4861 ARP / IPv6 Neighbor Discovery	Defines neighbor discovery for link-layer reachability.	→	DR-04 gate: **ping failure** alone never means **"gateway down"**.
RFC 7567 / 8290 AQM, FQ-CoDel	Describes modern queue management to control bufferbloat.	→	DR-03 gays **"consistent with queueing"**, never names the culprit.
RFC 2680 / 2681 / 3393 IPPM loss, delay, delay variation	Standard metrics for loss, delay, and delay variation.	→	**Matched windows with p95 and jitter** definitions.
RFC 8305 Happy Eyeballs v2	Specifies parallel IPv4/IPv6 connection strategy.	→	**Per-address-family probing** (IPv4 vs IPv6).
RFC 8484 / 8908 DoH, captive-portal API	Defines DNS over HTTPS and captive-portal detection mechanisms.	→	Add **limitation flags** on findings (DoH, captive portal).

exisiting work and capability comparision

Capability	Speed tests (Ookla, M-Lab NDT)	Test suites (Netalyzr [8])	Manual (ping, mtr, dig)	NERVE
Measures symptoms	✓	✓	✓	✓
Separates gateway vs external	—	Partial	Partial (expert only)	✓
Explains with a stated rule	—	—	—	✓
Picks next test adaptively	—	—	—	✓
Says "Inconclusive"	—	—	—	✓
Verifies the fix	—	—	—	✓

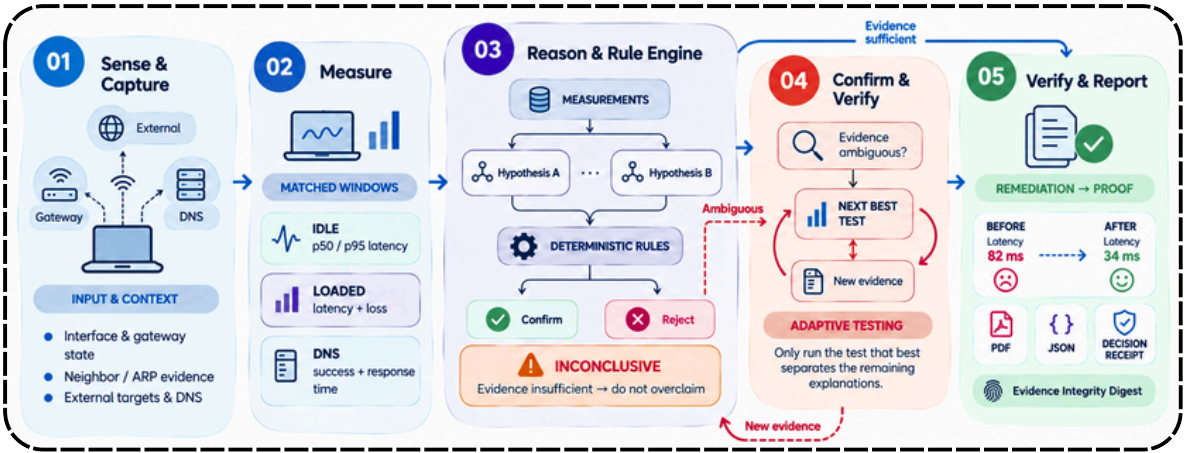
REFERENCES:

[1] RFC 2308 — DNS Negative Caching [2] RFC 826 — Address Resolution Protocol (ARP) [3] RFC 4861 — IPv6 Neighbor Discovery [4] RFC 7567 — Active Queue Management Recommendations [5] RFC 8290 — FQ-CoDel Queue Management [6] RFC 2680 — One-Way Packet Loss Metric [7] RFC 2681 — Round-Trip Delay Metric [8] RFC 3393 — IP Packet Delay Variation [9] RFC 8305 — Happy Eyeballs Version 2 [10] RFC 8484 — DNS over HTTPS (DoH)

To our knowledge, none of these combine adaptive test selection, explicit inconclusive and verified remediation. That's NERVE'S gap .

Proposed Solution

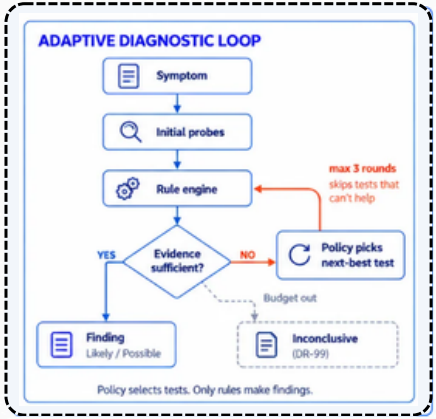
An evidence-driven diagnostic pipeline that separates network symptoms from defensible findings.



Optimization Proposed

v2.0 → v2.2: from 'run every test' to 'test only what the evidence demands'.

	BEFORE (v2.0)		AFTER (v2.2)	IMPACT
	Run every test, fixed order	›	Pick next test that splits hypotheses	Fewer tests
	Ping fails = gateway down	›	Needs ARP/neighbor + TCP fallback	No false 'router dead'
	Rate cap only	›	Rate cap + competing traffic	Queue really builds
	Random public subdomains	›	Controlled zone nerve.lab	No NXDOMAIN noise
	Root-cause, ISP fault, signed	›	Observation wording + integrity digest	Honest claims
	One baseline	›	3 baselines + CIs + ablations	Hard to cherry-pick



22	3	100%	0
refinements v2.0 → v2.2	baselines (B1, B2, B3)	replay determinism target	unsupported ISP/root-cause claims

Solution Architecture & Design

Policy chooses tests. Collectors measure. Only rules diagnose.

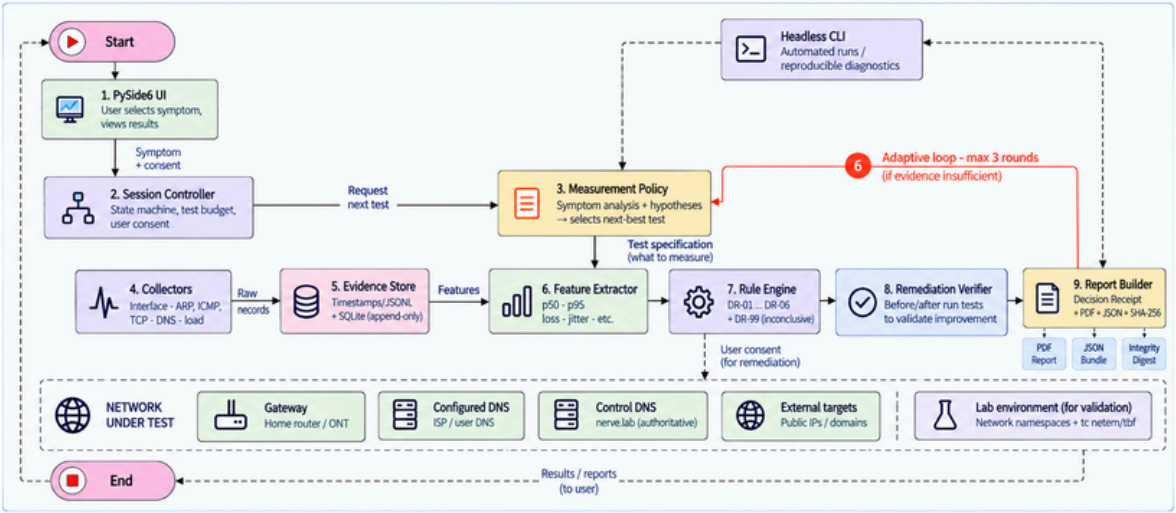
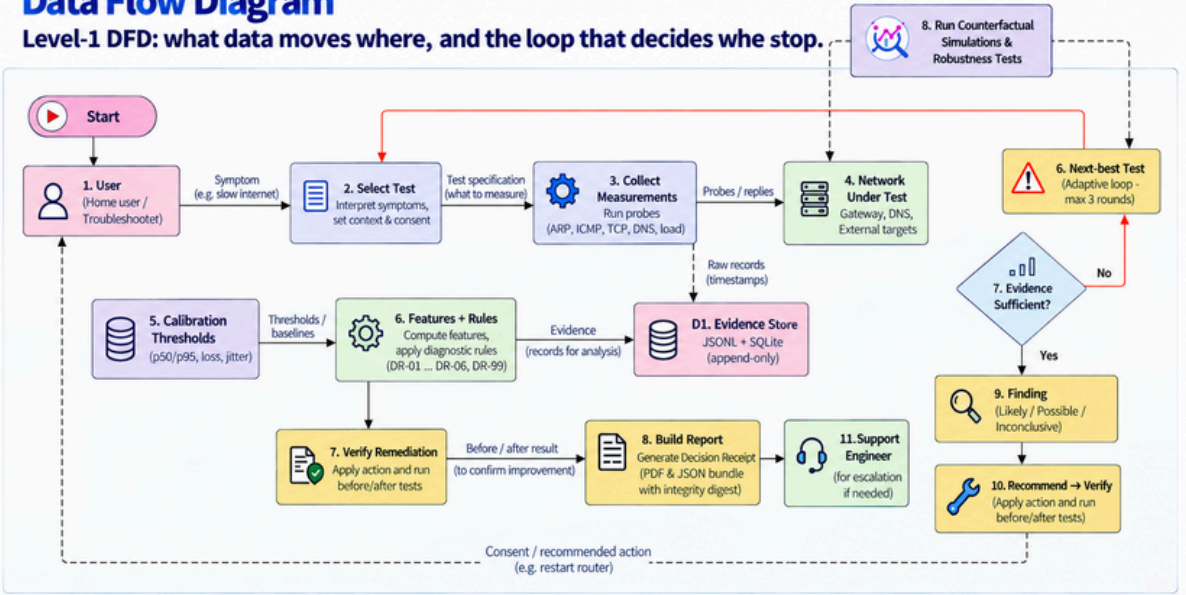


Figure: NERVE system architecture (UML component view)

Data Flow Diagram

Level-1 DFD: what data moves where, and the loop that decides whe stop.



NERVE — Deployment Timeline

From development to deployment: a phased plan to deliver a robust, reproducible MVP.

DEPLOYMENT PLAN (Proposed)

Target: Linux (Ubuntu 22.04/24.04) · Python 3.11+ · Local execution (no cloud)

Phase	Timeline	October 2026		November 2026		December 2026
		14 – 20 Oct	21 – 31 Oct	1 – 15 Nov	16 – 23 Nov	1 – 16 Dec
Phase 1	**Foundation**	14 – 20 Oct				
Phase 2	**Core engine**		21 – 31 Oct			
Phase 3	**Product + evaluation**			1 – 15 Nov		
Phase 4	**Validate + freeze**				16 – 23 Nov	
Phase 5	**Finals polish**					1 – 16 Dec

- **G1: collector fixture tests pass** — Environment setup, gateway discovery, RTT/loss/DNS collectors, JSON evidence schema, CLI.
- **G2: Tier 1 scenario tests pass** — Feature extraction, deterministic rules (DR-01 to DR-06 + DR-99), adaptive policy, baselines, replay.
- **G3: criteria evaluated** — Fault-injection lab, adaptive test selection, scenario tests, initial reports (PDF/JSON), Diagnostic Decision Receipt.
- **G4: feature freeze** — Calibration on real network runs, repeated trials, robustness checks, scope freeze (Tier 3 features cut first if needed).
- **G5: release tag** — UI/report polish, demo rehearsal, final documentation, reproducible evidence bundle, release tag.

METHODOLOGY & KEY NOTES

Phase gates
A tier starts only when the previous one passes its tests.

Pre-registered criteria
Pass targets committed before any trial is run.

Replay-based reproducibility
Same evidence file must reproduce the identical finding.

Scope freeze
Tier 3 is cut first if the schedule slips.

Bibliography / Standards Foundation

Core RFC standards and empirical literature underpinning the NERVE engine design.

IETF Standards & RFCs

Define the protocol behaviour, metrics and mechanisms used by NERVE.

[1] **RFC 2308**

Negative Caching of DNS Queries (DNS NCACHE).

Used for establishing the controlled nerve.lab zone to avoid NXDOMAIN negative-caching noise.

[2] **RFC 826 / RFC 4861**

ARP and IPv6 Neighbor Discovery.

Powers the DR-04 local gateway reachability gate, ensuring a failed ping alone never triggers a false "router dead" conclusion.

[3] **RFC 7567 / RFC 8290**

Active Queue Management (AQM), FQ-CoDel & BBR.

Forms the baseline for bufferbloat analysis, mapping load-related RTT inflation without assigning unfair blame.

[4] **RFC 2680 / RFC 2681 / RFC 3393**

IPPM Metrics for Packet Loss, Delay, and Jitter.

Defines the matched window summary statistics (p50, p95) used in NERVE's measurements.

Empirical Literature & Studies

Provide measurement insights and real-world methodology.

[5] **Gettys & Nichols (2011)**

"Bufferbloat: Dark Buffers in the Internet." Communications of the ACM, 54(1), 58–65.

Foundation for our idle vs. loaded latency experiment design.

[6] **Sundaresan et al. (2011 / 2016)**

"Broadband Internet Performance: A View from the Gateway." IMC 2011.

"Home Network or Access Link? A Longitudinal Study of Broadband Performance." PAM 2016.

Methodology for separating local gateway bottlenecks from last-mile and upstream access links.

Thank You